Applied Regression Interpretation

Gov 51: Data Analysis and Politics

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The Big Picture

Women earn 84 cents for every dollar men earn

Where does that number come from?

And what does it actually mean?

Regression turns data into claims you can interrogate

- **Today:** One dataset, three models, building complexity
- **Skills you need:**
  - Interpret every coefficient in context
  - Compute $\hat{Y}$ by plug-in
  - Compare models side-by-side
  - Read interactions

The Current Population Survey is the gold standard for labor data

- **What:** Monthly survey of ∼60,000 U.S. households
- **Who runs it:** Bureau of Labor Statistics + Census Bureau
- **How:** Interviewers visit or call sampled households
- **Why it exists:** Official source for unemployment rate, earnings, labor force participation
- Every jobs report you see in the news comes from the CPS

Wooldridge extracted 526 workers from the 1976 CPS

- Jeff Wooldridge compiled this sample for his econometrics textbook
- 1976: pre-Equal Pay enforcement, large raw gender gaps
- Clean, real data — every number we compute today is a fact about these 526 people

Variable	Type	Description
`wage`	Continuous	Hourly wage ($/hr)
`educ`	Continuous	Years of education
`exper`	Continuous	Years of work experience
`tenure`	Continuous	Years at current job
`female`	Binary (0/1)	= 1 if woman
`married`	Binary (0/1)	= 1 if married

$n = 526$ · 274 men, 252 women

Model 1: The Raw Gap

Model 1 starts with the population regression

$$\text{wage}_i \quad = \quad \beta_0 \; + \; \beta_1 \; \cdot \; \text{female}_i \; + \; \epsilon_i$$

- wage_i — the **outcome** (*dependent variable*): hourly wage for person i
- female_i — the **predictor** (*independent variable*): equals 1 if woman, 0 if man
- β_0, β_1 — **unknown parameters** we want to learn
- ϵ_i — the **error term**: everything else that affects wages

We estimate the unknowns using OLS

Recall: OLS chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the sum of squared residuals:

$$\min_{\hat{\beta}_0, \hat{\beta}_1} \sum_{i=1}^{n} \hat{e}_i^2 \quad = \quad \sum_{i=1}^{n} \left(Y_i \; - \; \hat{\beta}_0 \; - \; \hat{\beta}_1 X_i\right)^2$$

This gives us the **fitted values**:

$$\widehat{\text{wage}}_i \quad = \quad \hat{\beta}_0 \; + \; \hat{\beta}_1 \; \cdot \; \text{female}_i$$

Women earn $2.51 less per hour than men on average

	Model 1
Intercept ($\hat{\beta}_0$)	7.10
	(0.21)
Female ($\hat{\beta}_1$)	**−2.51**
	(0.30)
R^2	0.116
n	526

Five numbers. By the end of today, you will know what every one of them means.

What do 7.10 and −2.51 mean?

$$\widehat{\text{wage}}_i = \underbrace{7.10}_{\hat{\beta}_0} + \underbrace{(-2.51)}_{\hat{\beta}_1} \cdot \text{female}_i$$

If female = 0 (men), what is $\hat{Y}$?

If female = 1 (women), what is $\hat{Y}$?

So what is $\hat{\beta}_1$ measuring?

$\hat{\beta}_0$ is the male mean; $\hat{\beta}_1$ is the gap

- Men (female = 0): $\hat{Y} = 7.10 \quad \Rightarrow \quad \hat{\beta}_0 = \overline{\text{wage}}_{\text{men}}$
- Women (female = 1): $\hat{Y} = 7.10 - 2.51 = 4.59 \quad \Rightarrow \quad \hat{\beta}_0 + \hat{\beta}_1 = \overline{\text{wage}}_{\text{women}}$
- Subtract: $\hat{\beta}_1 = 4.59 - 7.10 = -2.51$

With binary X: intercept = baseline mean, slope = difference in means

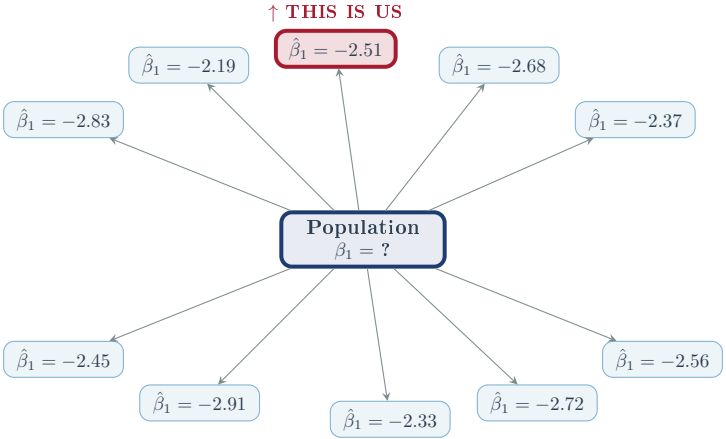
What is the (0.30) underneath the coefficient?

	Model 1
Female ($\hat{\beta}_1$)	-2.51
	(0.30) ← **what is this?**

We got -2.51 from 526 workers.

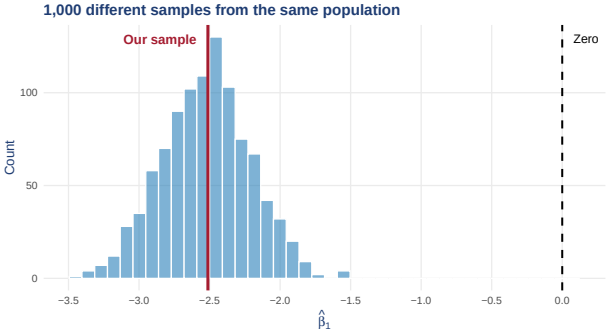
What if we had surveyed a *different* 526 workers?

Our sample is one of many we could have drawn



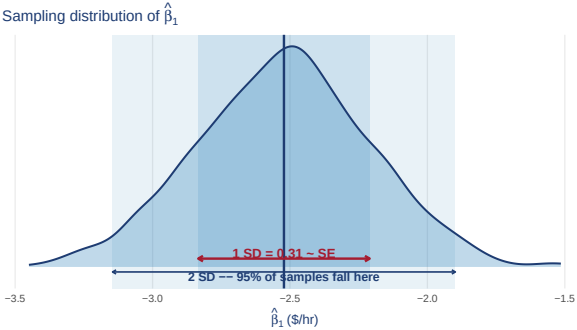
Every sample gives a different $\hat{\beta}_1$ — all are estimates of the same β_1

Here are 1,000 different samples from the same population



The **standard error** measures the width of this distribution: SE = 0.30

The SE is the standard deviation of this distribution



$$\text{SE}(\hat{\beta}_1) \quad \approx \quad \text{SD of all possible } \hat{\beta}_1\text{'s} \quad \approx \quad 0.30$$

β_1 is the truth; $\hat{\beta}_1$ is our guess from data

The population (all U.S. workers)

$\text{wage}_i = \beta_0 + \beta_1 \cdot \text{female}_i + \epsilon_i$

- β_1 = true average wage gap
- This is what we **want to know**
- Not a causal claim — just a difference

Our sample (526 workers)

$\hat{\beta}_1 = -2.51$

- One draw from that distribution
- From *this particular* sample
- The SE tells us how much it bounces

Hats mean "estimated from this sample" — not the true population value

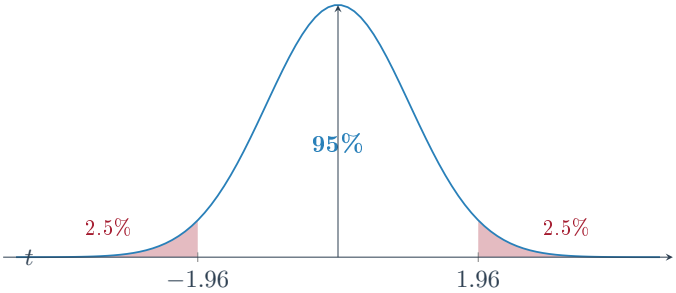
The t-statistic asks: could the true gap be zero?

- **Thought experiment:** Suppose $\beta_1 = 0$ (no real wage gap in the population)
- We got $\hat{\beta}_1 = -2.51$ — how surprising is that?
- Measure surprise in units of SE:

$$t = \frac{\hat{\beta}_1}{\mathrm{SE}(\hat{\beta}_1)} = \frac{-2.51}{0.30} = -8.28$$

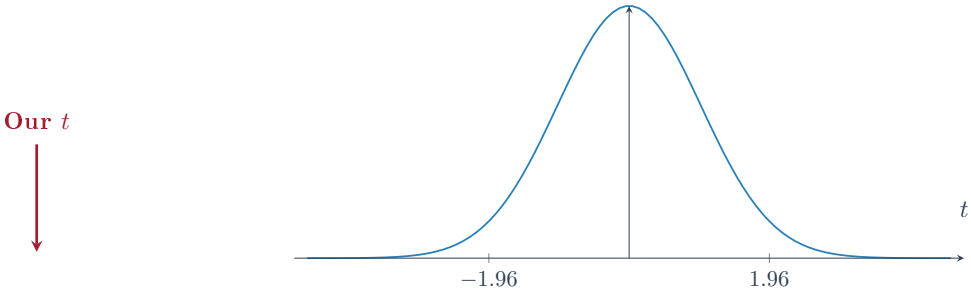
Our estimate is **8.28 standard errors** away from zero

The t-distribution shows which values are plausible under H_0



- Blue: 95% of t-statistics fall between ± 1.96 if $\beta_1 = 0$
- Red tails: only 5% land here — these are "unlikely under H_0"

Our $t = -8.28$ is not even on the chart



The p-value is the chance of landing this far out if $\beta_1 = 0$ — here it is ≈ 0

$p < 0.001$ means: if the true gap were zero, you'd essentially never see $t = -8.28$

From hypothesis test to confidence interval

- We rejected $\beta_1 = 0$ — zero is implausible
- But which values of β_1 **are** plausible?
- **Answer:** values we *cannot* reject — within 1.96 SEs of $\hat{\beta}_1$

$$95\% \text{ CI} \;=\; \hat{\beta}_1 \pm 1.96 \times \text{SE} \;=\; -2.51 \pm 1.96(0.30)$$

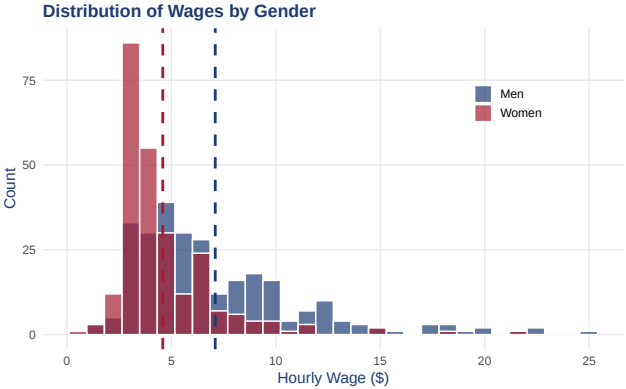
$$95\% \text{ CI} \quad = \quad (-3.10,\ -1.92)$$

We are 95% confident the true gap is between $-\$3.10$ and $-\$1.92$

- **Correct:** If we repeated this sampling 100 times, ∼95 of those intervals would contain β_1
- **Wrong:** "There is a 95% chance β_1 is in this interval"
  - β_1 is fixed — it's either in there or it isn't
- Zero is **not** in $(-3.10,\ -1.92)$ — consistent with rejecting H_0

You will need **1.96** for confidence intervals on the exam

Is knowing someone's gender enough to predict their wage?



Look at the overlap. How much wage variation does gender explain?

$R^2 = 0.116$ means gender alone explains 12% of wage variation

- 88% of variation is *within* gender groups
- Some men earn very little; some women earn a lot
- Gender matters, but it is far from the whole story

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\sum \hat{e}_i^2}{\sum (Y_i - \bar{Y})^2} = 0.116$$

Turn to your neighbor

"The wage gap is $2.51."

What does that number **not** tell you?

Model 2: Adding Covariates

Model 2 adds education, experience, and tenure

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{educ}_i + \hat{\beta}_3 \text{exper}_i + \hat{\beta}_4 \text{tenure}_i$$

- Same outcome, same treatment variable
- Now we **hold constant** education, experience, and tenure
- This asks a *different question* than Model 1

Before we look: will the female coefficient get bigger or smaller?

Model 1: $\hat{\beta}_{\text{female}} = -2.51$

We are about to add education, experience, and tenure.

	Men	**Women**
Mean education	12.8 yrs	12.3 yrs
Mean experience	17.6 yrs	16.4 yrs
Mean tenure	6.5 yrs	3.6 yrs

Will the magnitude of $\hat{\beta}_{\text{female}}$ increase or decrease? Why?

The gender gap shrinks from \$2.51 to \$1.81 after adding controls

	Model 1	Model 2
Intercept	7.10	−1.57
	(0.21)	(0.72)
Female	−2.51	**−1.81**
	(0.30)	(0.26)
Education		0.57
		(0.05)
Experience		0.03
		(0.01)
Tenure		0.14
		(0.02)
R^2	0.116	**0.364**
n	526	526

Every coefficient has a "holding constant" interpretation

- **Female:** $-\$1.81$/hr, holding education, experience, and tenure constant
- **Education:** $+\$0.57$/hr per additional year, holding gender, experience, and tenure constant
- **Experience:** $+\$0.03$/hr per additional year, holding gender, education, and tenure constant
- **Tenure:** $+\$0.14$/hr per additional year, holding gender, education, and experience constant

"Holding constant" changes the question we are asking

Model 1

$\hat{\beta}_{\text{female}} = -2.51$

Gap between the *average* man and the *average* woman

Model 2

$\hat{\beta}_{\text{female}} = -1.81$

Gap between a man and woman with the *same* education, experience, and tenure

Adding covariates changes what $\hat{\beta}_1$ means

R^2 tripled — the model explains 3× more variation

- Model 1: $R^2 = 0.116$
- Model 2: $R^2 = 0.364$
- Education, experience, and tenure explain a lot of wage variation
- The SE on Female *shrank*: $0.30 \rightarrow 0.26$
  - More precise estimate when the model fits better

Turn to your neighbor: plug-in prediction

Using Model 2:

$$\hat{Y} = -1.57 + (-1.81)\text{female} + 0.57 \cdot \text{educ} + 0.03 \cdot \text{exper} + 0.14 \cdot \text{tenure}$$

Predict hourly wage for:

Profile: educ = 16, exper = 10, tenure = 5

1. A woman with this profile: $\hat{Y} = ?$

2. A man with this profile: $\hat{Y} = ?$

3. What is the gap?

Plug-in prediction: the gap is always $\hat{\beta}_{\text{female}}$

- **Man** (female $= 0$):
  $\hat{Y} = -1.57 + 0(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$8.54$
- **Woman** (female $= 1$):
  $\hat{Y} = -1.57 + 1(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$6.72$
- **Gap**: $6.72 - 8.54 = -\$1.81 = \hat{\beta}_{\text{female}}$

With no interaction, the gap is the same for every profile

Model 3: Interactions

Does marriage affect everyone's wages the same way?

Think about this:

- In 1976, who was more likely to be the primary earner in a married household?
- If employers perceive married men as "more stable," what happens to their wages?
- If employers perceive married women as "likely to leave," what happens?

*How would you **model** this — one slope for marriage, or two?*

Model 3 asks: does marriage affect wages differently for men and women?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{married}_i + \hat{\beta}_3 (\text{female} \times \text{married})_i + \cdots$$

- $\hat{\beta}_2$ = effect of marriage **for men** (when female = 0)
- $\hat{\beta}_2 + \hat{\beta}_3$ = effect of marriage **for women**
- $\hat{\beta}_3$ = *difference* in marriage effect across genders

Marriage boosts men's wages by \$1.82 but *reduces* women's

	Model 1	Model 2	Model 3
Intercept	7.10	−1.57	−2.39
	(0.21)	(0.72)	(0.73)
Female	−2.51	−1.81	−0.31
	(0.30)	(0.26)	(0.41)
Married			1.82
			(0.40)
Female × Married			**−2.40**
			(0.53)
Education		0.57	0.55
		(0.05)	(0.05)
Experience		0.03	0.02
		(0.01)	(0.01)
Tenure		0.14	0.13
		(0.02)	(0.02)
R^2	0.116	0.364	**0.392**
n	526	526	526

The marriage premium: \$1.82 for men, −\$0.57 for women

Men (female = 0)

Marriage effect = $\hat{\beta}_2$
= +\$1.82/hr

Married men earn significantly more

Women (female = 1)

Marriage effect = $\hat{\beta}_2 + \hat{\beta}_3$
= $1.82 + (-2.40) = -\$0.57$/hr

Married women earn slightly less

$\hat{\beta}_3 = -2.40$ captures the *difference* in marriage premiums

Turn to your neighbor: 2×2 predictions

Using Model 3 with educ = 12, exper = 10, tenure = 5:

$$\hat{Y} = -2.39 + (-0.31)\text{female} + 1.82 \cdot \text{married} + (-2.40)\text{female} \times \text{married}$$
$$+0.55 \cdot \text{educ} + 0.02 \cdot \text{exper} + 0.13 \cdot \text{tenure}$$

Fill in the table:

	Single	**Married**
Men	?	?
Women	?	?

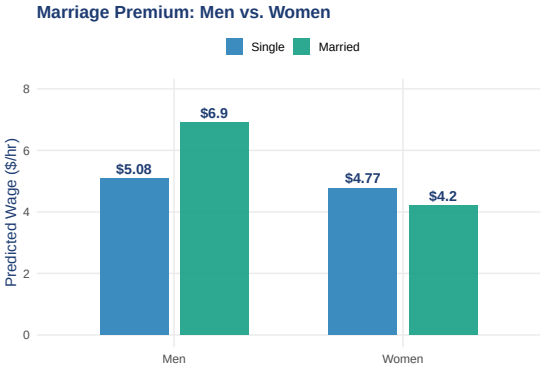
Which group earns the most? The least?

Married men earn the most; married women earn the least

	Single	Married
Men	$5.08	$6.90
Women	$4.77	$4.20

Marriage premium:

Men: +$1.82 Women: −$0.57



Putting It All Together

Track $\hat{\beta}_{\text{female}}$ across all three models

	Model 1	Model 2	Model 3
$\hat{\beta}_{\text{female}}$	-2.51	-1.81	-0.31
	(0.30)	(0.26)	(0.41)
Interpretation	Raw gap	Adjusted gap (same human capital)	Gap for single workers (same HC)
R^2	0.116	0.364	0.392

The "wage gap" depends on which question you ask

Four skills to master before the exam

1. **Interpret coefficients in context**
   "One more year of education $\rightarrow$ +\$0.57/hr, holding ... constant"
2. **Compute $\hat{Y}$ by plug-in**
   Substitute values into $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots$
3. **Compare models side-by-side**
   Track how coefficients, R^2, and SEs change
4. **Read interactions**
   $\hat{\beta}_3$ = difference in slopes $\rightarrow$ build the 2×2 table

Exam 1 is Thursday, March 12

- **Review guide:** posted on course website
- **Cheat sheet:** 2 pages, front and back, handwritten or typed
- **Practice:**
  - Plug-in predictions — get fast at arithmetic
  - Interpret every number in a regression table
  - Build 2×2 tables from interaction models

You have all the tools. Practice using them.